

ETEHGAN:End-to-End High-Capacity Generative Steganography Without Auxiliary Channel

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supervised

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1 GENERAL INTRODUCTION

- Context
- Types of Steganography
- Study of WYSAWIS
- Problem
- Project Objectives

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Consistent reports of data breaches in various sectors of life such as healthcare, finance, government, business, education and the increased use of social media has made data security an increasingly important issue today as many secure techniques are proposed by experts and many new threats are developed by attackers.

Data Security: The process of protecting digital information from unauthorized access, corruption, theft or loss throughout its life cycle with the aim of ensuring the confidentiality, integrity and availability of data (CIA). Cryptography and information hiding are the two data security techniques that have been used over the years to combat threats to the security of data.

Cryptography: The practice of securing communication and data by using coded algorithms to make information inaccessible to unauthorized parties. Cryptography Makes ciphertext difficult to decrypt

Information Hiding: The practice of protecting information by concealing it in another media or in another form to prevent unauthorized access. Information hiding makes stego image appear innocent and ordinary It involves **Steganography and Watermarking**

Diagrammatic illustration of the Domain of Information Hiding

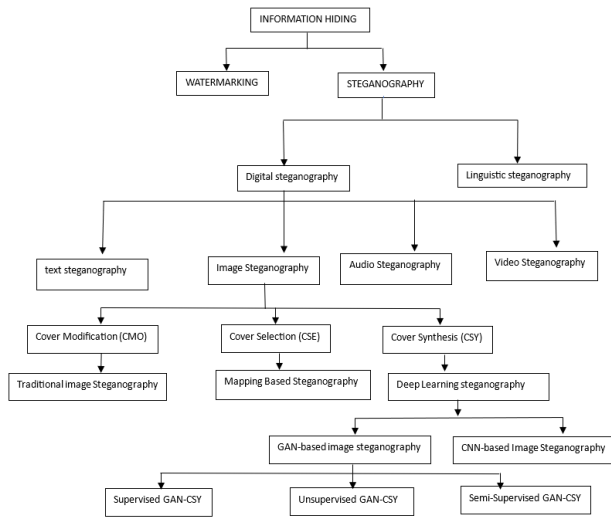


Figure 1: Information Hiding

Steganography: From Greek "steganos" (covered) + "graphia" (writing)

- Ancient art of concealing confidential information
- Media: pictures, sound, videos
- Secret format: audio, text, images and video

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1.Steganography by Cover Modification (CMO)

The Steganographer starts with a cover image and makes modifications to it in order to embed secret data. However, the modification will inevitably introduce some embedding changes into the cover image making it to look suspicious.CMO techniques include **Traditional Image Steganography** techniques which are:

Spatial Domain Methods:

- LSB (Least Significant Bit)
- HUGO(Highly Undetectable steGO)
- BOW(Bag of Words)
- UNIWARD(Universal Wavelet Relative Distortion)
- PVD(Pixel Value Differencing)

Transform Domain Methods:

- DWT (Discrete Wavelet Transform)
- DFT (Discrete Fourier Transform)
- DCT (Discrete Cosine Transform)
- IWT (Integer Wavelet Transform)

Challenges:

- Limited embedding capacity
- Modification of stego images making the stego image sensitive to steganalysis tools

Approach:

Steganography by cover selection is the process of strategically choosing an object (like an image, audio, or video file) to hide a secret message within. The goal is to select a cover media with characteristics that make it most suitable for hiding data, thereby maximizing embedding capacity while minimizing the visual or auditory distortion that would make the hidden information noticeable to an observer or steganalysis tools.

Limitations:

- Limited embedding capacity
- Dependence on auxiliary channel (e.g., shared cloud account)
- Redundancy management issues
- Atypical behavior could arouse suspicion

Steganography by cover synthesis is a modern technique that conceals secret data by generating a new image that looks identical to a natural image, rather than modifying an existing one.

Key Advantage:

- High embedding capacity
- Better security

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Rather than directly embed the secret message in pixels of an image, the new concept of coverless image steganography hides a message in a cover image without modifying it, which significantly increases the resilience to image steganalysis tools. Existing high-capacity coverless image steganography methods do not provide an efficient algorithm for rapidly generating a large number of possible hash sequences. Moreover, the sending of additional information generated during the hiding process and essential for extraction is not explicitly defined. This study overcomes the limits of building a large image data set and sending in order a large number of stego images by using a single image and a single cloud.

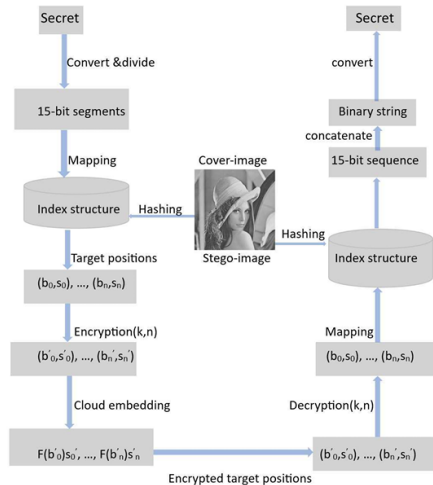


Figure 2: Architecture of WYSAWIS

WYSAWIS: High-Capacity Coverless Image Steganography

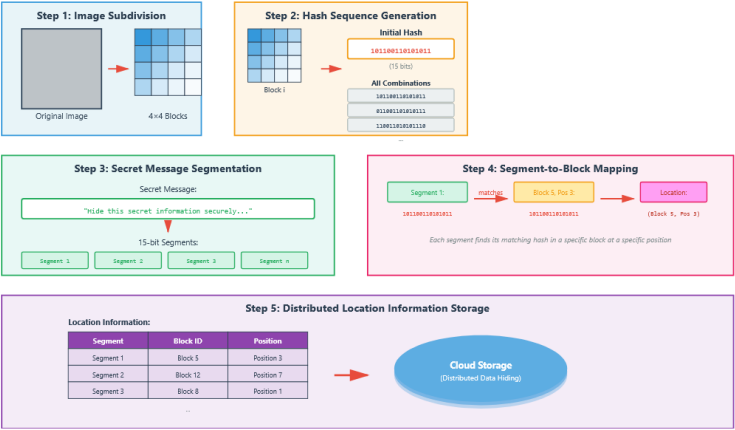


Figure 3: Information Hiding phase

- ➊ **Image Rank Determination:** After receiving the **stego-image**, the receiver searches for it in their identical image dataset (identical to the sender's) and determines its **rank** r .
- ➋ **Key Computation:** Using the rank r , the receiver computes the necessary **encryption keys**.
- ➌ **Stego-Folder Search:** The receiver logs onto the same cloud and searches for each of the files in the **stego-folder** within the **cover folders**, obtaining their cloud positions (folder number, file number).
- ➍ **Position Decryption:** These cloud positions (except the first one, which corresponds to the size of the message) are decrypted using the calculated encryption keys and corresponding algorithms, yielding the **target positions of the sequences** (block number, sequence number).
- ➎ **Hash Sequence Extraction:** Using the **stego-image**, the hash sequences corresponding to these target positions are calculated. All these sequences are **concatenated in order**.
- ➏ **Zero Removal/Message Extraction:** Depending on whether the size T (where $T = \text{folder number} \times 6435 + \text{file number}$) is a multiple of 15 or not, some zeros are removed or not to reveal the final **secret message**.

Method	Year	BER	Payload
WYSAWIS	2023	20.34	491520- ∞

Limitations:

- Dependence on auxiliary channel (e.g., shared cloud account)
- Redundancy management issues

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How to eliminate the dependency on auxiliary channel and redundancy management problem as seen in the reference publication WYSAWIS?

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The main objective is to develop a complete system(end-to-end) based on generative models (GANs) capable of synthesizing a realistic-looking steganographic image from a secret message, and extracting this message without any additional channel.

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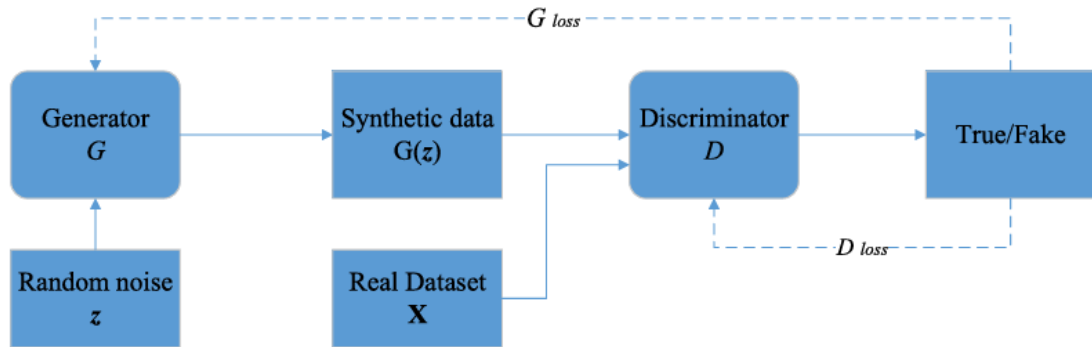


Figure 4: Image of a GAN by Goodfellow et al., 2014

Approach:

- Establish mapping rule between secret binary sequences and noise values
- Generate new image using DCGAN

Advantages:

- Significantly improved imperceptibility

Limitations:

- Secret cannot be perfectly extracted by training of extraction network
- Limited embedding capacity

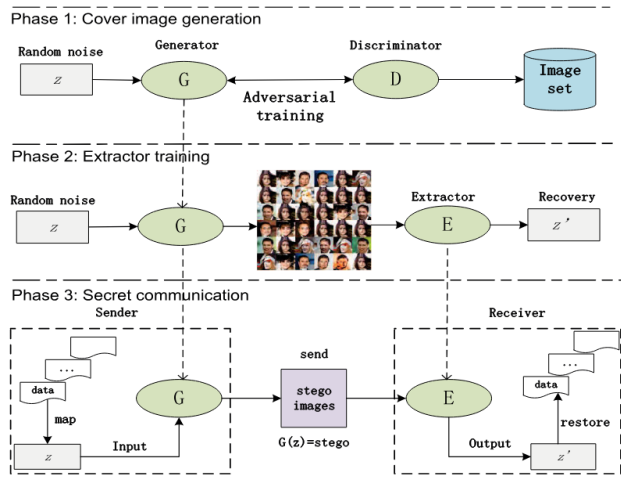


Figure 5: Architecture of DCGAN by Donghui Hu et al., 2018

Two Operations:

- **Encoding:** Takes cover image + binary message $\rightarrow$ creates stego image
- **Decoding:** Takes stego image $\rightarrow$ recovers binary message

Advantages:

- High embedding capacity (4.4 bpp)
- Increased perceptual quality

Limitations:

- Vulnerable to deep learning steganalysis tools

SteganoGAN Architecture

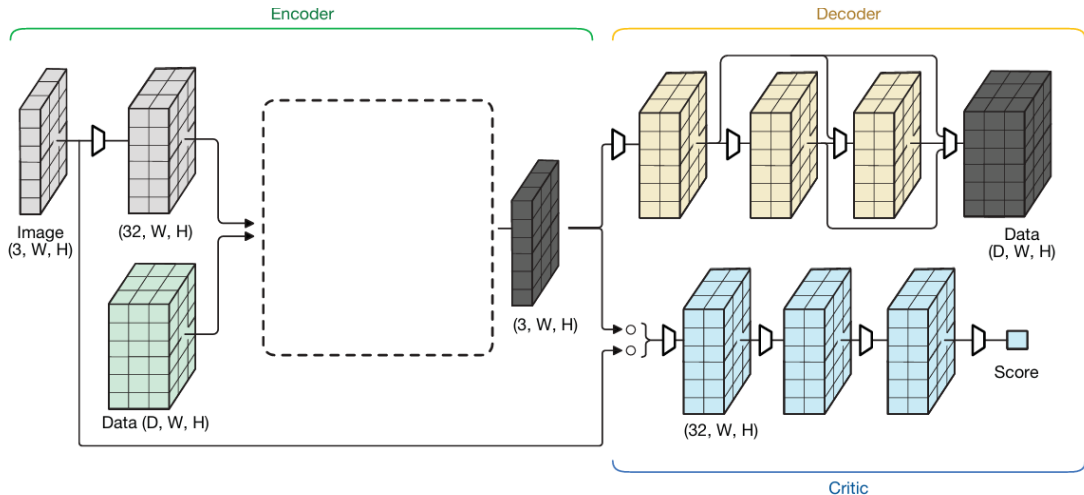


Figure 6: Architecture of SteganoGAN by Zhang et al., 2019

- Two-phase hiding:
 - **S1 Phase:** Mapping-based coverless steganography
 - **S2 Phase:** StarGAN generates stego image from facial attributes

Advantages:

- Significantly improved robustness and security

Limitations:

- Limited embedding capacity
- Image quality concerns

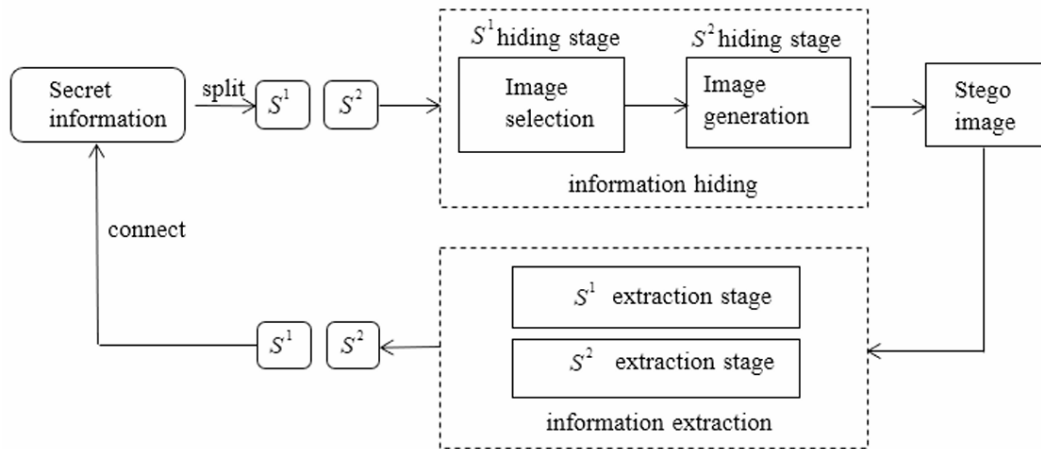


Figure 7: Architecture of StarGAN by Chen et al., 2021

Improved Training with WGAN-GP:

- Sender trains WGAN-GP with natural (disguise) image + secret image
- Generator G creates image visually identical to secret
- Generator parameters saved to form codebook
- Sender uploads disguise image to cloud
- Receiver downloads image and uses generator with parameters to obtain the secret image

Results:

- High image quality
- Good security

Limitations:

- Not universal - each secret needs new model
- Poor recovery quality due to lack of correlation

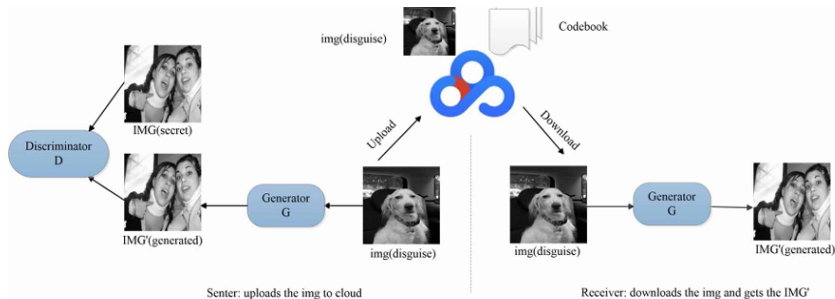


Figure 8: Architecture of WGAN-GP by Duan et al., 2021

Coverless Information Hiding with SRGAN:

- Map textual data to facial attribute labels of anime characters
- Generate anime characters based on these label at 128×128 , upscale to 512×512 using SRGAN

Extraction:

- Pre-trained CNN extracts facial attributes
- Reverse-map labels to recover hidden information

Advantages:

- Superior hiding capacity (1088 bpp)

Limitations:

- Needs more powerful GANs
- Better resistance to geometric attacks needed

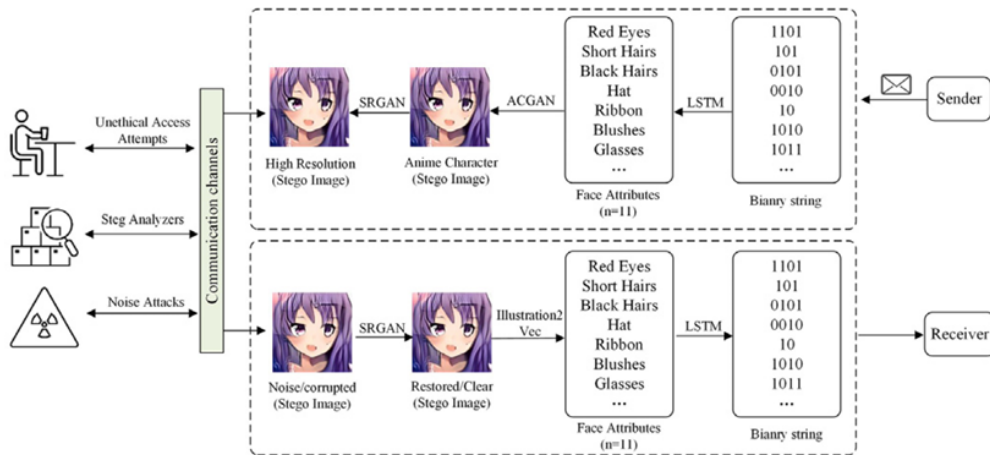


Figure 9: Architecture of SRGAN by Rehman et al., 2024

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1. Imperceptibility/Security:

- **PSNR (Peak Signal-to-Noise Ratio):** Measures quality by comparing pixel values. Higher values indicate better quality
- **SSIM (Structural Similarity Index):** Measures luminance, contrast, structure. Closer to 1 indicate greater similarity
- **FID (Fréchet Inception Distance):** Measures the visual quality and realism by comparing the statistical similarity of the generated image and the real image. Lower FID indicate higher realism
- **Entropy:** The randomness of the generated image. Higher values indicate more random pixels making harder to detect
- **Attack Success Rate:** Probability of successful steganalysis detection. Lower values indicate better security

2. Robustness:

- Withstanding non-geometric attacks (compression, noise)
- Withstanding geometric attacks (rotation, cropping)
- **BCR (Bit Correct Rate):** Ratio of correctly extracted bits. Higher values are better
- **BER (Bit Error Ratio):** Frequency of bit errors: Lower values are better

3. Capacity:

- Maximum information that can be hidden
- Expressed as bits per pixel (bpp)
- Higher bpp indicate that more data can be hidden

Imperceptibility Comparison

Method	Reference	PSNR	SSIM	FID	Entropy	Attack Success Rate%
DCGAN	DONGHUI HU et al. 2018	-	-	-	-	0.76%
SteganoGAN	Zhang et al. 2019	31.842	0.926	31.26	-	0.59%
StarGAN	Chen et al. 2020	-	-	-	1.50%	54.5%
WGAN-GP	Duan et al. 2021	35.1429	0.9457	-	-	-
CycleGAN	Koyoro et al. 2022	41.45	0.97	-	-	-%
SRGAN	Rehman et al. 2024	-	0.9999	-	-	1.56%
Adaptive-GAN	Polisetty et al. 2025	38.7	0.96	-	-	5%

Method	Reference	BER	BCR
DCGAN	DONGHUI HU et al. 2018	-	-
SteganoGAN	Zhang et al. 2019	0.261438	-
StarGAN	Chen et al. 2020	-	98.755%
WGAN-GP	Duan et al. 2021	-	-
SRGAN	Rehman et al. 2024	0.28978	-
Adaptive-GAN	Polisetty et al. 2025	1.2	-

Method	Reference	Image Size	Payload (bpp)
DCGAN	DONGHUI HU et al. 2018	64×64	0.07328
SteganoGAN	Zhang et al. 2019	256×256	4.4
StarGAN	Chen et al. 2020	128×128×3	>33 bits/carrier
WGAN-GP	Duan et al. 2021	256×256	1
CycleGAN	Koyoro et al. 2022	256×256×3	24.97
SRGAN	Rehman et al. 2024	8×8	1088

Method	Reference	Advantages	Disadvantages
DCGAN	DONGHUI HU et al. 2018	Improved security	Low embedding capacity and robustness
SteganoGAN	Zhang et al. 2019	Good embedding capacity and increased perceptual quality	Less security and robustness
StarGAN	Chen et al. 2020	Improved security and robustness	Limited embedding capacity and image quality
WGAN-GP	Duan et al. 2021	Good image quality and security	Low embedding capacity; each image requires a new model

Method	Reference	Advantages	Disadvantages
CycleGAN	Koyoro et al. 2022	High embedding capacity and security	Less robust against deep learning steganalysis tools
SRGAN	Rehman et al. 2024	Excellent embedding capacity and image quality	Less security and robustness against geometric attacks
Adaptive-GAN	Polisetty et al. 2025	Improved security and robustness	Low embedding capacity

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step 1: Image Subdivision

- Share and image database
- Divide the a single image into blocks 4x4 size

Step 2: Hash Sequence Generation

- For each 4×4 block:
- An initial 15-bit hash sequence is generated and all possible combinations/permutations of this sequence are computed creating a set of hash sequences for that block Example: Block 5 might have hash sequences like 101100110101011, 011001101010111, etc.

Step 3: Secret Message Segmentation

- The secret message is converted to binary
- The binary data is split into 15-bit segments
- Each segment needs to be mapped to a block

Step 4: Matching and Mapping

- Each 15-bit message segment is matched to a hash sequence from the available blocks
- When Segment 1 (101100110101011) matches a hash in
- Block 5 at Position 3, we record:
- Block ID: 5
- Sequence: 3 (which hash combination matched)
- This process repeats for all segments

step 5: Generation of stego image for Extraction of secret

- convert (Block ID, Sequence number....) into a binary sequence
- Give this binary sequence to a GAN to generate an image identical to the preshared image
- Transfer this image to the receiver

Step 6: Extraction of Secret Information

- The receiver gives this image to a GAN to obtain the binary sequence
- The receiver reconverts to decimal to obtain (Block ID, Sequence number)
- The receiver then obtains the corresponding hash sequences from the index structure using (Block ID, Sequence number)
- The receiver concatenates to obtain the secret message

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Project Objectives:

- Address channel dependency problem
- Eliminate redundancy management issues

Proposed Solution:

- Develop an **end-to-end generative steganography system**
- Completely eliminate the need for auxiliary channel

Expected Benefits:

- Higher security
- Better capacity
- No external channel dependencies

Thank You!

Questions?